Affine Divisibility Toeplitz Systems: Constructive Periods and Same-Base Pointed Factor Posets

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Abstract

Fix an integer $p \geq 3$ and a periodic directive u with exact finite support, least period at least two, and unequal cyclic neighbors. We study the pointed Toeplitz system generated by

$$x_{p,u}(k) = u_{\nu_p((p-1)k+1)}, \qquad k \in \mathbb{Z},$$

where ν_p is the p-divisibility exponent, also when p is composite. The affine coordinate has one exceptional residue modulo each p^N. We use it to determine the exact skeleton, prove that every p^N is essential, and exhibit the sequence as an aperiodic normal simple Toeplitz sequence. Under the all-coordinate finite-block convention, the pure-power period structure is constructive exactly when p is prime; every prime divisor ℓ of a composite p gives the strict common period ℓp^N at level N. Our factor classification is deliberately narrower: source and target lie in this affine family at the same fixed base, maps are onto and pointed, and targets are taken modulo pointed conjugacy. Within that scope, every onto pointed factor map is the unique coordinatewise extension of a surjective letter quotient. The mechanism is a sequence of high centers whose off-center local-rule windows are independent of the directive index. Consequently, factor targets are exactly the independent-set partitions of the directive's cyclic adjacency graph, ordered by refinement. Graphical Stirling numbers and the chromatic polynomial then enumerate the target classes. Deterministic finite checks are supplied only as reproducibility and falsification controls.

1 Introduction

The arithmetic family considered here turns a pointed factor problem into a finite quotient problem. For one integer base $p \geq 3$ and a periodic directive u, set

$$x_{p,u}(k) = u_{\nu_p((p-1)k+1)}. \tag{1}$$

The exponent in Equation (1) records divisibility by powers of p; it is not assumed to be a valuation in the ring-theoretic sense when p is composite. Nevertheless, the affine term $(p-1)k+1$ selects a single nested residue class at every power p^N. That one-hole geometry simultaneously controls periodic structure and supplies unusually rigid centers for sliding block codes.

Pointed homomorphisms of Toeplitz systems with a common period structure are generally described by maps between aligned symbols (Downarowicz et al., 1995). The question addressed here is more specific: when both systems arise from Equation (1) at one fixed base and the chosen Toeplitz points are preserved, can an aligned-symbol map retain genuine context dependence? The answer is no. At a center c_n of sufficiently high level, every bounded noncentral offset has a letter independent of n. A local rule therefore sees the directive index only through the center letter. Periodicity of the directives and density of the chosen orbit then promote this observation to the whole subshift.

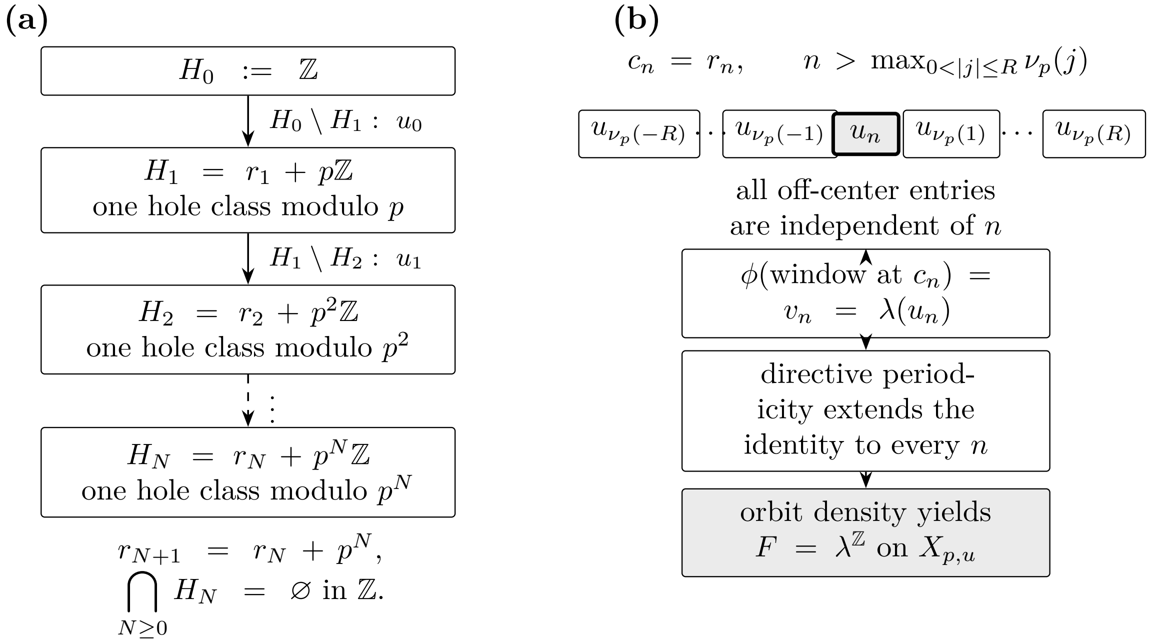

Figure 1: The same affine congruence drives both parts of the classification. (a) At level N, all coordinates outside the single class $H_N = r_N + p^N\mathbb{Z}$ are p^N-periodic, while the layer $H_N \setminus H_{N+1}$ is filled by u_N. (b) At a sufficiently high center $c_n = r_n$, a finite local-rule window depends on n only through its center letter. Pointedness gives $v_n = \lambda(u_n)$; directive periodicity and orbit density then force $F = \lambda^{\mathbb{Z}}$ on the entire source subshift.

The complete scope is important. All factor maps below are continuous, shift commuting, onto, and pointed. Source and target belong to the affine family at the same fixed base p; factor targets are classified modulo pointed conjugacy. Constructiveness uses the least common period of *all* positions in the initial finite block, following the terminology of Hosseini and Yassawi (2026) with that indexing convention stated explicitly. We do not classify different-base maps, nonpointed maps, arbitrary simple Toeplitz systems, or factors outside the affine target family.

Our four theorem-level contributions form one dependency chain.

(i) We calculate the exact p^N-skeleton, prove that each p^N is essential, and obtain an aperiodic normal simple Toeplitz realization for every integer $p \geq 3$.

(ii) We determine the constructive status of the initial blocks: constructiveness holds at every level exactly for prime p, while a prime divisor of composite p supplies a strict counterperiod.

(iii) Using the high-center identity, we prove that every onto pointed factor between same-base members of the family is the unique surjective coordinate letter quotient.

(iv) We identify pointed-conjugacy classes of factor targets with independent-set partitions of a finite cyclic adjacency graph. Factor arrows are refinement arrows, and graphical Stirling numbers enumerate the target classes.

The bounded computations reported later are reproducibility support, not a fifth theorem contribution and not evidence for an infinite quantifier.

Figure 1 displays the two mechanisms without replacing their proofs. The single-hole skeleton is valid for prime and composite bases. Only the finite-block constructive property splits by primality. By contrast, the high-center argument uses no modular inverse and so gives the same pointed factor rigidity at every integer base in scope.

The article is organized as follows. Section 2 records the nearest general factor criteria and the precise residual question. The family and all quantifiers are frozen in Section 3. Exact skeletons and the prime/composite split are proved in Section 4. High centers and factor rigidity are treated in Section 5; the finite quotient poset follows in Section 6. We close with exact examples, bounded proof diagnostics, and the remaining scope boundaries.

2 Toeplitz factor criteria and the residual question

The relevant earlier results separate into period-structure descriptions, factor criteria over the odometer zero, and obstructions for pure-power structures. The comparison in Table 1 is by hypotheses and quantifiers, not by priority.

Period structures and one-hole models. A Toeplitz sequence is built from coordinates that are periodic along full arithmetic progressions. Increasing essential periods organize those coordinates into skeletons. Such structures also enter Bratteli–Vershik models for Toeplitz flows (Gjerde and Johansen, 2000). General factor questions for Toeplitz and almost one-to-one extensions have a broader scope than the explicit family below (Downarowicz and Durand, 2002). Here the nested skeletons have one hole class and are calculated directly from one affine congruence; no general representation theorem is needed for that calculation.

Pointed maps over zero. For Toeplitz sequences with the same period structure, Downarowicz et al. (1995, Theorem 1) characterize homomorphisms over zero through a map between aligned level symbols. Their criterion is the nearest general owner of the factor problem considered here. We specialize to one explicit same-base family and ask when all such finite-level context collapses to a map of individual directive letters. Our direct proof uses the Curtis–Hedlund–Lyndon representation of shift-commuting continuous maps (Hedlund, 1969) and high centers, rather than translating the aligned symbol notation.

Constructive pure-power structures. Hosseini and Yassawi (2026) define constructive period structures through the essential period of an initial finite block and obtain cross-base necessary conditions for pointed factors between constructive pure-power Toeplitz shifts. In particular, their theorem gives a divisibility obstruction between the bases and equality of bases for pointed conjugacy. We use their terminology, state the all-coordinate block convention explicitly, and determine exactly which bases in Equation (1) meet it. Cross-base sufficiency is not considered. This finite-block constructiveness question is separate from the factor classification: the latter will be proved for every integer base in scope.

After these owner subtractions, the residual question is narrow: for the affine divisibility family at one fixed base, when does every onto pointed over-zero aligned-symbol factor with an in-family target collapse to one unique letter quotient? The answer will follow from a local normal form at the centers r_n. A bounded source search found no earlier theorem with this exact conjunction of object class, map quantifiers, all-radius collapse, and quotient-poset conclusion. We use that statement only to delimit the comparison; it is not a claim of exhaustive coverage, novelty, or priority.

Table 1: Scope of the two nearest general owners and the specialization proved here. "Over zero" and "pointed" in the map rows mean that the distinguished Toeplitz point is preserved. The present factor row concerns only onto maps at one fixed base with source and target both in the affine family; the present period row has no map quantifier. Cross-base sufficiency and nonpointed maps remain outside the scope.

Source	Period/base assumptions	Map quantifier	Conclusion used for owner subtraction
Downarowicz–Kwiatkowski–Lacroix (1995)	Same period structure	Homomorphism over zero	An aligned level-symbol map characterizes existence; bijectivity characterizes isomorphism.
Hosseini–Yassawi (2026)	Constructive pure-power structures (p^n) and (q^n)	Pointed factor or pointed conjugacy	Necessarily $q \mid p$ for a factor and $p = q$ for a conjugacy; no cross-base sufficiency is asserted.
Present specialization: periods	Any one integer base in the affine divisibility family	No map quantifier	Under the all-coordinate finite-block convention, constructiveness holds exactly for prime bases; composite bases have explicit strict common counterperiods.
Present specialization: factors	One fixed integer base; both systems in the affine divisibility family	Onto pointed map; target classes modulo pointed conjugacy	Independently of constructiveness, every such map collapses to the unique letter quotient, and target classes are admissible partitions.

3 The affine family and conventions

This section fixes the objects once, including the hypotheses that will be used for surjectivity, uniqueness, and quotient normalization.

Let $\mathcal{A}$ be a finite alphabet, let $\mathcal{A}^{\mathbb{Z}}$ carry the product topology, and write σ for the left shift. A subshift is a closed σ-invariant subset. For $x \in \mathcal{A}^{\mathbb{Z}}$ and $q \geq 1$, put

$$\mathrm{Per}_q(x) = \{k \in \mathbb{Z} : x(k + tq) = x(k) \text{ for every } t \in \mathbb{Z}\}. \tag{2}$$

The q-skeleton is the partial word that records x on $\mathrm{Per}_q(x)$ and places a hole elsewhere. A positive q is *essential* when this skeleton has least positive translation period q. A divisibility chain of essential periods whose periodic positions cover $\mathbb{Z}$ is a period structure.

We use *normal simple Toeplitz sequence* in the following local sense. There are nested periodic hole sets $H_0 \supset H_1 \supset \cdots$, with one residue class of holes at each stage, such that every layer $H_N \setminus H_{N+1}$ is filled uniformly by one letter. The presentation is *normal* when $\bigcap_N H_N = \varnothing$, so no integer coordinate remains unfilled. This is the only meaning of that term used below.

For a finite interval $I \subset \mathbb{Z}$, a common period of $x|_I$ means a positive q for which every $k \in I$ belongs to $\mathrm{Per}_q(x)$. Its essential period is the least such q. In particular, every coordinate in the interval is included. A pure-power period structure (p^N) is called *constructive* here when the initial block

$$B_N = x[0, p^N - 1] \tag{3}$$

has essential period p^{N+1} for every $N \geq 1$. This is the all-coordinate finite-block convention used throughout.

Definition 3.1 (Frozen directive)**.** A directive is a bi-infinite periodic sequence $u = (u_n)_{n\in\mathbb{Z}}$ over $\mathcal{A}$. It is *frozen* if its least period h satisfies $h \geq 2$, its exact support is $\mathcal{A} = \{u_0, \ldots, u_{h-1}\}$, and

$$u_n \neq u_{n+1} \qquad (n \in \mathbb{Z}). \tag{4}$$

Thus the last and first letters in a least-period word are also unequal.

Fix an integer $p \geq 3$. For a nonzero integer m, define

$$\nu_p(m) = \max\{e \geq 0 : p^e \mid m\}. \tag{5}$$

When p is composite, ν_p is only the p-divisibility exponent; we will never infer that every residue not divisible by p is a unit. Put

$$L_p(k) = (p-1)k + 1, \qquad r_N = 1 + p + \cdots + p^{N-1} = \frac{p^N - 1}{p-1} \quad (N \geq 1), \tag{6}$$

and set $r_0 = 0$. The equation $L_p(k) = 0$ has no integral solution because $p - 1 \geq 2$. Thus the following definition never evaluates Equation (5) at zero.

Definition 3.2 (Affine divisibility Toeplitz system)**.** For a frozen directive u, let

$$x_{p,u}(k) = u_{\nu_p(L_p(k))}, \qquad X_{p,u} = \mathrm{cl}\{\sigma^t x_{p,u} : t \in \mathbb{Z}\}, \tag{7}$$

and retain the distinguished point in $T_{p,u} = (X_{p,u}, \sigma, x_{p,u})$.

A morphism $F : T_{p,u} \to T_{p,v}$ means a continuous onto map commuting with the shifts and satisfying $F(x_{p,u}) = x_{p,v}$. Source and target always use the same fixed base p. We call such a map a pointed factor. A pointed conjugacy is an invertible pointed factor.

For a frozen directive on $\mathcal{A}$, define the simple graph

$$G_u = (\mathcal{A}, E_u), \qquad E_u = \{\{u_i, u_{i+1}\} : i \in \mathbb{Z}/h\mathbb{Z}\}. \tag{8}$$

A partition $\mathcal{P}$ of $\mathcal{A}$ is *admissible* if every block is an independent set of G_u. We say $\mathcal{P}$ *refines* $\mathcal{Q}$ if every $\mathcal{P}$-block is contained in a $\mathcal{Q}$-block. For an admissible $\mathcal{P}$, put $u_n^{\mathcal{P}} = [u_n]_{\mathcal{P}}$ and reduce this quotient directive to its least period; the canonical quotient target is $T_{\mathcal{P}} = T_{p,u^{\mathcal{P}}}$ on the block alphabet $\mathcal{A}/\mathcal{P}$.

The results can now be stated as a single theorem. Separate statements and proofs in the next sections expose the dependencies.

Theorem 3.3 (Main classification)**.** *Fix an integer $p \geq 3$ and frozen directives u and v.*

(a) For every $N \geq 1$,

$$\mathrm{Per}_{p^N}(x_{p,u}) = \mathbb{Z} \setminus (r_N + p^N\mathbb{Z}). \tag{9}$$

Every p^N is essential, $(p^N)_{N\geq 1}$ is a period structure, and $x_{p,u}$ is an aperiodic normal simple Toeplitz sequence.

(b) The structure is constructive in the convention of Equation (3) if and only if p is prime. If p is composite and ℓ is any prime divisor of p, then $\ell p^N < p^{N+1}$ is a common period of B_N for every $N \geq 1$.

(c) A pointed factor $T_{p,u} \to T_{p,v}$ exists if and only if there is a surjective letter map $\lambda : \mathcal{A} \to \mathcal{B}$ such that

$$v_n = \lambda(u_n) \qquad (n \in \mathbb{Z}). \tag{10}$$

In that case λ and the factor map are unique, and $F = \lambda^{\mathbb{Z}}$ on $X_{p,u}$. Pointed conjugacy is exactly a bijective relabeling of directive letters, with no phase shift.

(d) *For fixed $T_{p,u}$, pointed-conjugacy classes of same-base in-family factor targets are in bijection with the admissible partitions of $\mathcal{A}$. Before quotienting by pointed conjugacy, the canonical representatives form a thin category: there is a unique factor arrow $T_{\mathcal{P}} \to T_{\mathcal{Q}}$ exactly when $\mathcal{P}$ refines $\mathcal{Q}$. After identifying arbitrary relabelings with their canonical representatives, the class-level object is the admissible-partition refinement poset. If $S_{G_u}(k)$ counts its partitions into k nonempty independent blocks, then*

$$\#\{k\text{-letter target classes}\} = S_{G_u}(k), \tag{11}$$

$$\min\{\text{target alphabet size}\} = \chi(G_u), \tag{12}$$

$$P_{G_u}(z) = \sum_k S_{G_u}(k)(z)_k. \tag{13}$$

A binary target exists exactly when G_u is bipartite.

The proof graph is short but rigid. Coordinate periodicity gives the exact skeleton, which gives essential periods and the one-hole recursion. A separate common-period calculation gives the prime/composite split. The high-center identity and the Curtis–Hedlund–Lyndon theorem give the letter map, after which kernels become admissible graph partitions. No finite enumeration is used in any of these implications.

4 Skeletons and constructive periods

The affine term first gives periodicity at each coordinate, with an argument that remains valid for composite bases.

Lemma 4.1 (Coordinate periodicity)**.** *Let $k \in \mathbb{Z}$ and $e = \nu_p(L_p(k))$. Then p^{e+1} is a period of the position k in $x_{p,u}$.*

Proof. Write $L_p(k) = p^e a$ with $p \nmid a$. For every $t \in \mathbb{Z}$,

$$L_p(k + tp^{e+1}) = p^e\big[a + (p-1)tp\big]. \tag{14}$$

The bracket is congruent to a modulo p, so it is not divisible by p. The exponent of Equation (14) is therefore e, and the directive letter is unchanged along the whole arithmetic progression. No inverse modulo p was used. □

The centers in Equation (6) satisfy two identities that will recur:

$$L_p(r_N) = p^N, \qquad r_{N+1} = r_N + p^N. \tag{15}$$

Theorem 4.2 (Exact one-hole skeleton)**.** *For every $N \geq 1$,*

$$\mathrm{Per}_{p^N}(x_{p,u}) = \mathbb{Z} \setminus (r_N + p^N\mathbb{Z}).$$

Proof. Since $\gcd(p-1, p) = 1$, the congruence in Equation (6) gives

$$p^N \mid L_p(k) \iff k \equiv r_N \pmod{p^N}. \tag{16}$$

If k is outside this residue class, put $e = \nu_p(L_p(k)) < N$ and write $L_p(k) = p^e a$, where $p \nmid a$. Then, for every $t \in \mathbb{Z}$,

$$L_p(k + tp^N) = p^e\big[a + (p-1)tp^{N-e}\big]. \tag{17}$$

The bracket in Equation (17) is a modulo p; hence the exponent and letter remain constant. Thus every nonhole position lies in $\mathrm{Per}_{p^N}(x_{p,u})$.

Conversely, take any $k \equiv r_N \pmod{p^N}$. Its p^N-progression contains both r_N and $r_{N+1} = r_N + p^N$. By Equation (15), their letters are u_N and u_{N+1}. They are unequal by Equation (4), so no coordinate in the alleged hole progression is p^N-periodic. This proves the reverse inclusion and the equality. □

Corollary 4.3 (Essential periods and simple Toeplitz form)**.** *Every p^N is essential, the powers $(p^N)_{N\geq 1}$ form a period structure, and $x_{p,u}$ is an aperiodic normal simple Toeplitz sequence.*

Proof. The p^N-skeleton has exactly one hole residue modulo p^N. Any positive translation period of that partial word must preserve the hole set, and hence must be divisible by p^N. Its least period is therefore p^N.

Define

$$H_N = \{k : p^N \mid L_p(k)\} = r_N + p^N\mathbb{Z}, \qquad H_0 = \mathbb{Z}. \tag{18}$$

The layer $H_N \setminus H_{N+1}$ consists exactly of coordinates with exponent N, so it is filled uniformly by u_N, leaving the one hole class H_{N+1}. No integer lies in every H_N: otherwise the fixed nonzero integer $L_p(k)$ would be divisible by all powers of p. Together with lemma 4.1, this shows that the periodic positions cover $\mathbb{Z}$ and that the one-hole filling has no residual coordinate. This is the normal simple Toeplitz recursion.

If $x_{p,u}$ had a nonzero global period q, translation by q would preserve each exact skeleton and therefore each unique hole class in Equation (18). Hence $p^N \mid q$ for every N, impossible for a fixed nonzero integer. Thus the sequence is aperiodic. □

Skeleton essentiality does not by itself settle constructiveness: the latter concerns the common period of all positions in a finite block. The next theorem separates these notions.

Theorem 4.4 (Constructiveness is equivalent to primality)**.** *Let $B_N = x_{p,u}[0, p^N - 1]$. Its essential period equals p^{N+1} for every $N \geq 1$ if p is prime. If p is composite and ℓ is any prime divisor of p, then ℓp^N is a common period of B_N and satisfies $\ell p^N < p^{N+1}$.*

Proof. We first obtain a common upper period for every integer base. If $0 \leq k < p^N$, then

$$0 < L_p(k) < p^{N+1}.$$

Hence $e = \nu_p(L_p(k)) \leq N$. The calculation in lemma 4.1 shows that p^{N+1} is a period of every such position. Thus p^{N+1} is a common period of B_N.

Suppose now that p is prime and that a positive common period q is not divisible by p^{N+1}. Write

$$q = p^j d, \qquad 0 \leq j \leq N, \qquad p \nmid d. \tag{19}$$

Because $(p-1)d$ is invertible modulo p^2, there is $t \in \mathbb{Z}$ with

$$1 + (p-1)dt \equiv p \pmod{p^2}. \tag{20}$$

The coordinate r_j lies in $[0, p^N - 1]$, where $r_0 = 0$, and

$$\begin{aligned} L_p(r_j) &= p^j, \\ L_p(r_j + tq) &= p^j\big[1 + (p-1)dt\big]. \end{aligned}$$

By Equation (20), the two divisibility exponents are exactly j and $j+1$. Their letters u_j and u_{j+1} differ, contradicting that q is a period of the position r_j. Thus every common period is divisible by p^{N+1}, and the upper bound gives equality.

Finally let p be composite, fix a prime $\ell \mid p$, and put $q = \ell p^N$. Since $\ell < p$, this period is strictly smaller than p^{N+1}. Fix $0 \leq k < p^N$, let $e = \nu_p(L_p(k))$, and take an arbitrary $t \in \mathbb{Z}$. If $e < N$, write $L_p(k) = p^e a$ with $p \nmid a$. Then

$$L_p(k + tq) = p^e\big[a + (p-1)t\ell p^{N-e}\big]. \tag{21}$$

The bracket is a modulo p, so the exponent remains e. If $e = N$, Equation (16) and $0 \le k < p^N$ force $k = r_N$, and

$$L_p(k + tq) = p^N\left[1 + (p-1)\ell t\right]. \tag{22}$$

The bracket is 1 modulo ℓ, so it cannot be divisible by p. Again the exponent is unchanged. Since t was arbitrary, ℓp^N is a full-progression period of every position of B_N. $\square$

The proof covers $p = 3$ and every composite base without treating a nonmultiple of p as a unit outside the prime lane. If one adopts a literal finite-block convention omitting coordinate zero, the same split holds; the replacement witness is recorded in Appendix A.

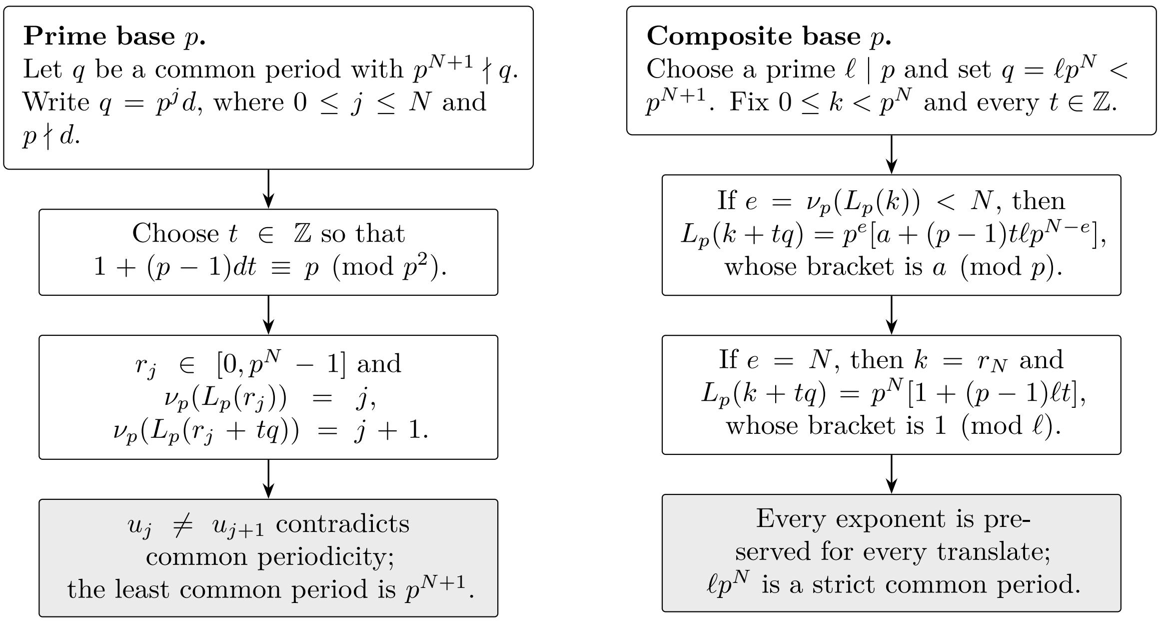

Figure 2: Schematic proof map for the finite-block dichotomy. In the prime lane, every candidate common period missing a factor of p^{N+1} is contradicted at a center r_j by a congruence modulo p^2. In the composite lane, each prime divisor ℓ of p supplies the strict common period ℓp^N; the exponent-N coordinate is checked separately. The theorem quantifies over every integer translate, not only the displayed witnesses.

5 High centers and pointed factor rigidity

The exact skeleton furnishes centers at which every bounded off-center coordinate has stabilized. This is the step that collapses arbitrary finite context to a letter map.

Lemma 5.1 (High-center identity)**.** *Let $c_n = r_n$. For every nonzero $j \in \mathbb{Z}$ and every $n > \nu_p(j)$,*

$$\nu_p(L_p(c_n + j)) = \nu_p(j). \tag{23}$$

Proof. Put $e = \nu_p(j)$ and write $j = p^e a$, where $p \nmid a$; the integer a may be negative. By Equation (15),

$$L_p(c_n + j) = p^n + (p-1)j = p^e\big[p^{n-e} + (p-1)a\big]. \tag{24}$$

Since $n - e \geq 1$, the bracket is congruent to $(p-1)a \equiv -a$ modulo p. It is not divisible by p, proving Equation (23). The calculation uses no inverse and applies to prime and composite bases alike. □

We use the Curtis–Hedlund–Lyndon theorem in the following standard form: a continuous shift-commuting map between subshifts is induced by a finite local rule (Hedlund, 1969). Padding its window if necessary, we may take a symmetric radius R.

Theorem 5.2 (Arbitrary-radius pointed factors collapse to letters)**.** *Let u and v be frozen directives at the same base p. A pointed factor $F : T_{p,u} \to T_{p,v}$ exists if and only if there is a surjective letter map $\lambda : \mathcal{A} \to \mathcal{B}$ satisfying $v_n = \lambda(u_n)$ for all $n \in \mathbb{Z}$. When it exists, λ is unique and*

$$F(z)(k) = \lambda(z(k)) \qquad (z \in X_{p,u},\ k \in \mathbb{Z}). \tag{25}$$

In particular, the pointed factor itself is unique.

Proof. Suppose first that F is a pointed factor. Choose $R \geq 0$ and a local rule ϕ such that

$$F(z)(k) = \phi(z[k-R, k+R]). \tag{26}$$

If $R > 0$, set

$$M = \max\{\nu_p(j) : 0 < |j| \leq R\}; \tag{27}$$

if $R = 0$, put $M = -1$. For every $n > M$, the center coordinate is $x_{p,u}(c_n) = u_n$, while lemma 5.1 gives

$$x_{p,u}(c_n + j) = u_{\nu_p(j)} \qquad (0 < |j| \leq R). \tag{28}$$

Thus all entries in the radius-R window except its center are independent of n. Every source letter occurs at arbitrarily large indices because u is periodic with exact support. For $a \in \mathcal{A}$, let W_a be the stabilized window with center a and define

$$\lambda(a) = \phi(W_a). \tag{29}$$

This is independent of the chosen large occurrence of a.

Pointedness and $L_p(c_n) = p^n$ now yield, for $n > M$,

$$v_n = x_{p,v}(c_n) = F(x_{p,u})(c_n) = \lambda(u_n). \tag{30}$$

Let h_u and h_v be the least directive periods and set $H = \operatorname{lcm}(h_u, h_v)$. Given any $n \in \mathbb{Z}$, choose s so large that $n + sH > M$. Periodicity extends Equation (30) to

$$v_n = v_{n+sH} = \lambda(u_{n+sH}) = \lambda(u_n). \tag{31}$$

Therefore, at every $k \in \mathbb{Z}$,

$$x_{p,v}(k) = v_{\nu_p(L_p(k))} = \lambda\big(u_{\nu_p(L_p(k))}\big) = \lambda(x_{p,u}(k)). \tag{32}$$

The original map F and $\lambda^{\mathbb{Z}}$ agree on the distinguished point, hence on its shift orbit by equivariance, and hence on its closure by continuity. This proves Equation (25).

Every target letter is some v_n by exact target support. Equation (31) therefore makes λ surjective. Exact source support also makes it unique: if $u_n = a$, then necessarily $\lambda(a) = v_n$. Consequently F is unique as well.

Conversely, suppose that a surjective letter map satisfies $v = \lambda \circ u$. Its coordinatewise extension is continuous and shift commuting and sends $x_{p,u}$ to $x_{p,v}$. We verify its image by two inclusions. If $y \in X_{p,u}$, choose integers n_i such that $\sigma^{n_i} x_{p,u} \to y$. Continuity and equivariance give

$$\lambda^{\mathbb{Z}}(y) = \lim_i \sigma^{n_i} x_{p,v} \in X_{p,v},$$

so $\lambda^{\mathbb{Z}}(X_{p,u}) \subseteq X_{p,v}$. Conversely, $\lambda^{\mathbb{Z}}(X_{p,u})$ is compact and therefore closed, and it contains $\mathrm{Orb}(x_{p,v})$ because it contains $x_{p,v}$ and is shift invariant. It consequently contains $\mathrm{cl}\,\mathrm{Orb}(x_{p,v}) = X_{p,v}$. The two inclusions prove equality, so the coordinatewise extension is onto and is the required pointed factor. □

Corollary 5.3 (Pointed conjugacy)**.** *Two systems $T_{p,u}$ and $T_{p,v}$ are pointedly conjugate exactly when their directives differ by a bijective relabeling of letters. The conjugacy has no phase shift.*

Proof. A bijective letter map has the inverse coordinate map. Conversely, the inverse of a pointed conjugacy is again pointed and same-base. Apply theorem 5.2 in both directions. The two resulting letter maps compose to the identity on every directive letter, because every such letter occurs at a center c_n. They are therefore bijections. □

Remark 5.4 (Where the argument uses the scope)**.** Same-base scope provides common centers c_n in Equation (30); pointedness identifies the local-rule output there with v_n rather than a shifted target coordinate; and an in-family target supplies its directive and exact support. Removing any of these hypotheses breaks a displayed step. The theorem does not assert that wrong-base or nonpointed maps are impossible in broader categories.

6 The pointed factor poset

Once theorem 5.2 has removed all local context, factor targets are controlled by kernels of finite letter maps. Cyclic adjacency is the only obstruction to remaining inside the frozen directive family.

For a partition $\mathcal{P}$ of $\mathcal{A}$, let $\pi_{\mathcal{P}} : \mathcal{A} \to \mathcal{A}/\mathcal{P}$ send a letter to its block and write

$$u_n^{\mathcal{P}} = \pi_{\mathcal{P}}(u_n). \tag{33}$$

When this directive has a shorter period than u, it is always normalized to its least period. Write $T_{\mathcal{P}} = T_{p,u^{\mathcal{P}}}$ for this canonical quotient target; its alphabet is literally the block set $\mathcal{A}/\mathcal{P}$.

Theorem 6.1 (Admissible partitions classify factor targets)**.** *Fix $T_{p,u}$. The following sets are in bijection:*

(i) pointed-conjugacy classes of same-base factor targets inside the affine family;

(ii) partitions of $\mathcal{A}$ whose blocks are independent in G_u.

At the representative level, there is a factor $T_{\mathcal{P}} \to T_{\mathcal{Q}}$ exactly when $\mathcal{P}$ refines $\mathcal{Q}$, and this factor is unique. Thus the canonical quotient targets form a thin category. Passing to pointed-conjugacy classes identifies arbitrary relabelings with these canonical representatives and gives the admissible-partition refinement poset.

Proof. By theorem 5.2, a factor is a surjective letter map $\lambda : \mathcal{A} \to \mathcal{B}$ satisfying $v = \lambda \circ u$. Its kernel partition $\mathcal{P}$ has the property

$$\lambda(u_i) \neq \lambda(u_{i+1}) \quad \iff \quad \text{no edge } \{u_i, u_{i+1}\} \text{ lies in one } \mathcal{P}\text{-block.} \tag{34}$$

Thus the image directive has unequal cyclic neighbors exactly when every block of $\mathcal{P}$ is independent in G_u. Exact source support implies that every block occurs in $u^{\mathcal{P}}$. Neighbor inequality excludes period one, and reduction to the least period preserves exact support and neighbor inequality. Hence every factor kernel is admissible and gives a frozen in-family target.

Conversely, if $\mathcal{P}$ is admissible, the block map $\pi_{\mathcal{P}}$ is surjective and Equation (33) is a frozen directive after least-period reduction. The converse direction of theorem 5.2 supplies the pointed factor.

It remains to identify target classes rather than labeled block maps. Two factor maps with the same kernel induce a well-defined bijection between their image alphabets, sending the image of each source letter under one map to its image under the other. Their image directives therefore differ by a bijective relabeling, so corollary 5.3 makes the targets pointedly conjugate. Conversely, a pointed conjugacy between two targets is a bijective relabeling. Equality at every directive index then says that two source letters have the same image under the first factor exactly when they have the same image under the second. Their kernels are equal.

Finally, a factor from the canonical $\mathcal{P}$-quotient to the canonical $\mathcal{Q}$-quotient is, by theorem 5.2, forced on every source letter to satisfy $\mu([a]_{\mathcal{P}}) = [a]_{\mathcal{Q}}$. This formula is well defined exactly when every $\mathcal{P}$-block lies in one $\mathcal{Q}$-block, which is precisely refinement; in that case it also constructs the factor. Exact support makes this block map unique. Hence the canonical targets form the asserted thin category. The preceding kernel argument then gives the asserted refinement poset only after passage to pointed-conjugacy classes. □

The word *poset* is exact here. We make no general lattice claim: the meet or join of two set partitions can leave the admissible subposet.

Corollary 6.2 (Graphical enumeration)**.** *Let $S_{G_u}(k)$ denote the number of partitions of $V(G_u)$ into exactly k nonempty independent blocks. Then $S_{G_u}(k)$ is the number of k-letter target classes modulo pointed conjugacy. The minimum target-alphabet size is $\chi(G_u)$, and a binary target exists exactly when G_u is bipartite. Moreover,*

$$P_{G_u}(z) = \sum_k S_{G_u}(k)(z)_k, \qquad (z)_k = z(z-1)\cdots(z-k+1). \tag{35}$$

Proof. The first assertion is theorem 6.1. The smallest number of independent blocks in a vertex partition is the chromatic number. The graph has at least one edge by cyclic neighbor inequality, so $\chi(G_u) \geq 2$; consequently, a two-block partition exists exactly when the graph is bipartite.

To count proper colorings by z labeled colors, first choose the partition into k nonempty color classes and then injectively assign k color labels to the blocks. There are $(z)_k$ such assignments. Summing over k proves Equation (35). □

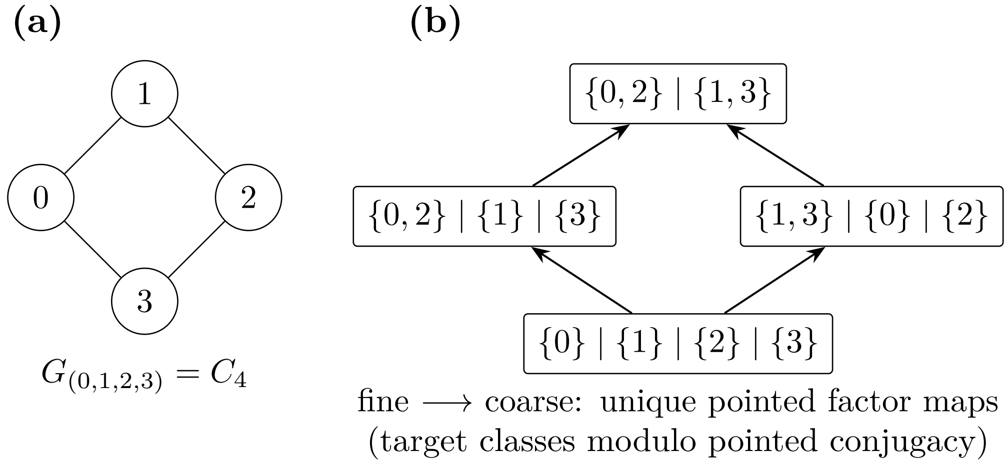

Figure 3: For the directive $(0,1,2,3)$, the adjacency graph is the four-cycle. Its admissible partitions are exactly the four displayed independent-set partitions; they represent pointed-conjugacy classes of same-base targets inside the frozen affine family. Upward edges are unique pointed factor maps and correspond to refinement from fine to coarse. This example is a diamond; the general classification is only a refinement poset, with no lattice assertion.

Example 6.3 (The four-cycle)**.** For $u = (0,1,2,3)^{\mathbb{Z}}$, the graph G_u is C_4. Its admissible partitions are

$$\begin{aligned}&\{0\} \mid \{1\} \mid \{2\} \mid \{3\},\\ &\{0,2\} \mid \{1\} \mid \{3\}, \qquad \{1,3\} \mid \{0\} \mid \{2\},\\ &\{0,2\} \mid \{1,3\}.\end{aligned}$$

Hence $S_{C_4}(2) = 1$, $S_{C_4}(3) = 2$, and $S_{C_4}(4) = 1$, so

$$P_{C_4}(z) = (z)_2 + 2(z)_3 + (z)_4.$$

7 Exact examples and proof diagnostics

Three small examples separate the notions used above before we report the bounded falsification controls.

Example 7.1 (Prime base)**.** Let $p = 3$. Then

$$r_1 = 1, \qquad r_2 = 4, \qquad r_3 = 13,$$

so the first hole classes are $1 + 3\mathbb{Z}$, $4 + 9\mathbb{Z}$, and $13 + 27\mathbb{Z}$. They are exact skeleton holes for every frozen directive. The initial block of length 3^N has essential period 3^{N+1}; in particular, the three coordinates of B_1 share period 9 and no smaller positive common period.

Example 7.2 (Composite base)**.** Let $p = 4$. The exact skeleton periods are still 4^N, with hole residues $r_1 = 1$, $r_2 = 5$, and $r_3 = 21$. However, choosing the prime divisor $\ell = 2$ in theorem 4.4 gives the common period

$$2 \cdot 4^N < 4^{N+1}$$

for every initial block B_N. At $N = 1$, the strict counterperiod is $8 < 16$. This example prevents one from conflating an essential skeleton period with the common period of a finite collection of positions.

Table 2: Deterministic finite falsification controls reproduced from the frozen candidate and audit receipts. The implementations use distinct evaluators, and all typed mutations are rejected. These bounded counts are not used to prove any theorem.

Control	Stage-2	Reciprocal	Root
Formula / nested-hole comparisons	68,288	24,024	–
Skeleton checks	28,764	1,519	984,576
Prime smaller periods rejected	3,918	19,766	200,988
Composite progression / shift checks	998,025	80,563	426,088
High-center identities	1,920	7,200	3,744
Partition tests	477	1,632	9,874
Bounded local-rule cases	972	–	–
Typed mutation controls	4	4	–
Local-rule false positives / negatives	0/0	–	–

Finite falsification controls only. Lock prefixes: candidate `c070bd76d8a2`, reciprocal `8b4d54a8cc7c`, root `e1dca456aecf`.

Example 7.3 (Least-period reduction after quotienting)**.** The directive $(0,1,0,2)^{\mathbb{Z}}$ is frozen. Merging the nonadjacent letters 1 and 2 is admissible and produces $(0,a,0,a)^{\mathbb{Z}}$, whose least period is two rather than four. This is why quotient directives are normalized after applying the block map. By contrast, merging 0 with 1 identifies an adjacency edge and leaves the frozen target family.

7.1 Finite falsification controls

The proof above is deductive and independent of computation. We nevertheless used deterministic finite searches to catch indexing errors, improper composite-base inversions, wrong refinement directions, and unlicensed scope extensions. Table 2 reports selected counts from three frozen checkers. The Stage-2 implementation compares the direct affine formula with a nested-hole evaluator and includes a bounded local-rule census. A reciprocal checker independently implements its arithmetic, directive, partition, and chromatic routines. A third checker uses larger skeleton, constructiveness, and partition ranges. None imports the theorem proof as code.

The local-rule census tested 972 bounded source/target/radius/base cases; its 132 context-consistent cases were exactly its 132 letter-quotient cases, with no false positive or false negative. This equality motivated a search for a uniform proof but is not used in theorem 5.2. Typed mutations include a wrong-base identity proposal, a nonpointed shift, an adjacent merge, a nonsurjective target map, and the false all-base constructive assertion. Rejection of a wrong-base or nonpointed mutation means only that it lies outside the frozen contract, not that no such map exists in a larger category.

The full deterministic protocol, input hashes, and rerun command are given in Appendix B. All integer counts in Table 2 are exact for the declared finite ranges, and every infinite statement is closed separately in Sections 4 to 6.

8 Scope and conclusion

The affine coordinate $(p-1)k+1$ links three levels of structure. Its power divisibility isolates one hole class modulo p^N; its behavior on a finite interval separates prime from composite bases under the constructive period convention; and its translates around r_n freeze every bounded off-center window. The last fact converts the general finite-local-rule description into a unique letter quotient. Once that collapse is known, the fixed-source factor problem becomes the refinement poset of independent-set partitions of a finite graph.

All conclusions retain the hypotheses declared in Section 3. The base is one fixed integer $p \geq 3$, the directives are periodic with exact support and unequal cyclic neighbors, maps are onto and pointed, and both source and target lie in the affine family. The target classification is modulo pointed conjugacy. We have not classified maps between different bases, maps over a nonzero odometer element, nonpointed maps, arbitrary simple Toeplitz systems, or factors outside the family. The admissible partition order is a poset; the results do not promote it to a lattice in general.

Two directions remain natural within these boundaries. One is to replace pointedness by a prescribed nonzero odometer phase and determine whether a translated version of the high-center normal form survives. The other is to combine the prime-base constructive obstruction with additional arithmetic data to study cross-base sufficiency. Neither question is settled by the same-base classification proved here.

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A Boundary conventions and alternate indexing

This appendix records the edge cases that can otherwise hide in the compact statements of theorems 4.4, 5.2 and 6.1.

A.1 If coordinate zero is omitted from the finite block

Our convention includes all coordinates $0, \dots, p^N - 1$. An alternate literal reading of a displayed finite-word definition in the source of the constructive terminology omits coordinate zero. The prime/composite split is unchanged under that reading.

Only the case $j = 0$ in the prime lower-bound proof uses $r_0 = 0$. Suppose p is prime and the alleged common period q is not divisible by p. The coordinate $k = 1$ belongs to every block with $N \geq 1$ and satisfies $L_p(1) = p$. Choose $s \in \mathbb{Z}$ such that

$$1 + (p-1)qs \equiv p \pmod{p^2} \tag{36}$$

and use translation multiplier $t = ps$. Then

$$L_p(1 + tq) = p\big[1 + (p-1)qs\big].$$

The exponents at 1 and $1 + tq$ are exactly 1 and 2, so the letters u_1 and u_2 differ. For $j \geq 1$, the original center r_j is nonzero and the proof in theorem 4.4 applies unchanged. The composite counterperiod proof quantifies over every retained coordinate and therefore also survives omission of zero.

A.2 Radius zero and negative offsets

For a radius-zero local rule, the set in Equation (27) is empty and we define $M = -1$. Every nonnegative center index then lies above the threshold, and the proof of theorem 5.2 immediately defines λ from the central letter. No separate compactness argument is needed beyond the orbit-closure step already given.

Negative offsets cause no sign issue. The exponent $\nu_p(j)$ is defined by divisibility of the nonzero integer j, and in Equation (24) one writes $j = p^e a$ with possibly negative a. The congruence $-a \not\equiv 0 \pmod{p}$ remains valid.

A.3 Support, neighbor inequality, and normalization

Exact support has two logically distinct uses. Source support lets us define and identify $\lambda(a)$ from a directive occurrence of every $a \in \mathcal{A}$; target support makes that letter map surjective. Neighbor inequality is used to exclude an alleged skeleton hole by the pair u_N, u_{N+1} and to keep quotient directives outside the period-one case. After a quotient, reduction to the least directive period is essential, as example 7.3 shows, but it does not remove any block letter from the infinite directive's support.

A.4 Composite divisibility language

Every argument outside the prime lane uses only the implication "not divisible by p implies divisibility exponent zero." It never uses the false composite-base implication that such a residue is invertible modulo p. The only modular inverse in the article occurs in Equation (20), where p is explicitly prime.

B Deterministic verification protocol

The finite checks in Section 7.1 are an audit trail for proof development. They are deliberately separated from the logical proofs of unbounded statements.

B.1 Two arithmetic evaluators

The first evaluator computes $\nu_p((p-1)k+1)$ directly by repeated exact division. The second starts with one hole class, fills $H_N \setminus H_{N+1}$ by u_N, and iterates until the queried coordinate leaves the next hole. The implementations do not import one another. A third, auditor-owned checker independently implements both viewpoints as well as restricted-growth enumeration of directives and set partitions.

For constructiveness, the finite searches quantify over all tested coordinates and all tested integer progression multipliers, rather than comparing only one translated block. Prime-base searches return an exact congruence witness for each smaller candidate period; composite-base searches check the strict period ℓp^N in both direct and nested-fill evaluators. Graph checks enumerate independent-set partitions, refinement pairs, and the falling-factorial expansion of the chromatic polynomial.

B.2 Frozen receipts

The writer-owned diagnostic receipt binds the following SHA-256 values:

Artifact	SHA-256
Stage-2 self-excluding manifest	`c070bd76d8a28e1b918fa040d9346db3` `2776f238e7081d8c3504648b137a583e`
Reciprocal-audit self-excluding manifest	`8b4d54a8cc7ce505dea3c26110346b31` `5af0e30c8753e5c95227d6ff82d75ca3`
Root-audit result	`e1dca456aecfe25a7c1a133d33c1a6a6` `bf43724775c0a4a3b5c4a1199e32ee64`

The generated table input has SHA-256

```
d0d8627e8becaee43a0eccecff206293
812e6f8fceed445e5f5efc657448fb05,
```

and the resulting LaTeX bytes have SHA-256

```
73562e6763d2df25826aa3fea56f5d014
02a5539c0570ca4887e1bb128b5ba5c.
```

Two isolated regenerations were byte-identical.

B.3 Typed mutations

The control suite includes the following intended failures.

1. A declared target letter omitted by a letter map is rejected as nonsurjective.
2. Merging a cyclic adjacency edge is rejected because the quotient leaves the frozen target family.
3. A base-3 to base-4 identity proposal is rejected after a distinguished-point mismatch. This is a scope control, not a theorem that all wrong-base factors are impossible.
4. A shift map is rejected as nonpointed after a distinguished-point mismatch, again without a broader impossibility claim.

5. At $p = 4$ and $N = 1$, the period $8 < 16$ rejects the false statement that every integer base has a constructive period structure.

B.4 One-command writer check

From the root of the reproducibility archive, run

```
PYTHONDONTWRITEBYTECODE=1 python3 qa/run_writer_checks.py
```

The script rehashes the three frozen receipts without writing to them, regenerates Table 2 in an isolated temporary directory, independently enumerates all 15 partitions of the four-cycle's vertex set, and checks that exactly four are admissible with four Hasse covers. A passing finite rerun increases confidence that the packaged statements were transcribed correctly; it does not establish any theorem quantified over all bases, levels, radii, or directives.